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143 ,FaceMaskify: A Deep Learning Approach to Face Mask Classification

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Abstract

- Face mask detection has become essential for global health, especially after recent respiratory pandemics.
- This work uses the **Face Mask Detection dataset by Omkar Gurav**, consisting of thousands of labeled images of masked and unmasked faces.
- A **deep learning Convolutional Neural Network (CNN)** model is developed for real-time mask detection.
- The model accurately classifies faces as **masked** or **unmasked**.
- It performs well under various conditions, including:
 - Different lighting conditions
 - Various face orientations
 - Complex backgrounds
- Data preprocessing and augmentation techniques** are applied to improve model performance and robustness.
- Transfer learning** is used to enhance accuracy and reduce training time.
- The proposed model shows **competitive performance** compared to existing methods in terms of accuracy and efficiency.
- The system can be integrated into **smart surveillance systems and edge devices**.
- Overall, the model helps improve **public safety and health monitoring** in real-world environments.

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Introduction

- After the COVID-19 pandemic, wearing face masks became essential to prevent the spread of respiratory diseases.
- Ensuring proper mask usage in public places is important for maintaining public health and safety.
- Manual monitoring of mask compliance is difficult, time-consuming, and not efficient.
- There is a need for an **automated system** to detect whether people are wearing masks.
- This project proposes a **face mask detection system using Convolutional Neural Networks (CNN)**.
- The system can **analyze images and detect mask usage in real-time**.
- It works under different conditions such as varying lighting, face angles, and backgrounds.
- Novelty of the project:**
 - Uses deep learning for accurate detection
 - Supports real-time monitoring
 - Can be integrated with surveillance systems and edge devices

Literature Survey

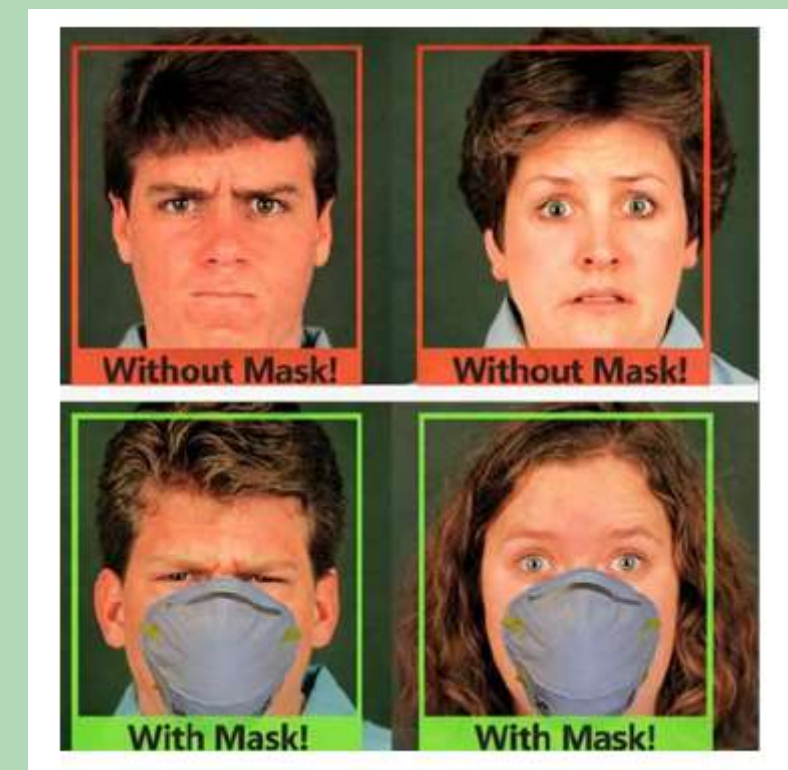
- Recent advancements in face mask detection are mainly driven by **Convolutional Neural Networks (CNNs)** and real-time object detection frameworks.
- Al Amin et al. proposed a CNN-based system capable of accurately distinguishing masked and unmasked faces with good computational efficiency.
- Himeur et al. explored deep learning and transfer learning techniques for mask detection in smart city environments, showing adaptability to real-world surveillance data.
- Dewi et al. used **YOLOv8 architecture** to improve detection accuracy and speed, making it suitable for real-time applications.
- Wei et al. enhanced YOLOv8 with context-aware convolution to handle occlusion and lighting variations.
- Tabassum et al. introduced a parallel CNN architecture for better feature extraction in crowded environments.
- Reddy et al. analyzed multiple models and highlighted trade-offs between accuracy and inference time.
- Ismael developed a real-time mask detection system suitable for live video surveillance.
- Abbas and Rathod summarized existing approaches and identified challenges like dataset diversity and real-time performance.
- Mamatha et al. proposed a CNN-based system with reliable results in controlled environments.
- Himel and Islam used **EfficientNet** to improve accuracy while reducing computational cost.
- Gao et al. optimized YOLO models for better performance in real-world conditions.
- Mostafa et al. proposed a YOLO-based system for drone surveillance in large public areas.
- Pandey et al. developed hybrid deep learning models to improve detection accuracy in complex environments.

Objectives

- Develop an **automatic face mask detection system** using deep learning models such as CNN and MobileNetV2.
- Classify faces into two categories:
 - With Mask
 - Without Mask
- Achieve **high accuracy and efficiency** in detecting mask usage.
- Enable **real-time monitoring** using camera systems.
- Ensure the system works under different conditions:
 - Lighting variations
 - Face orientations
 - Complex backgrounds

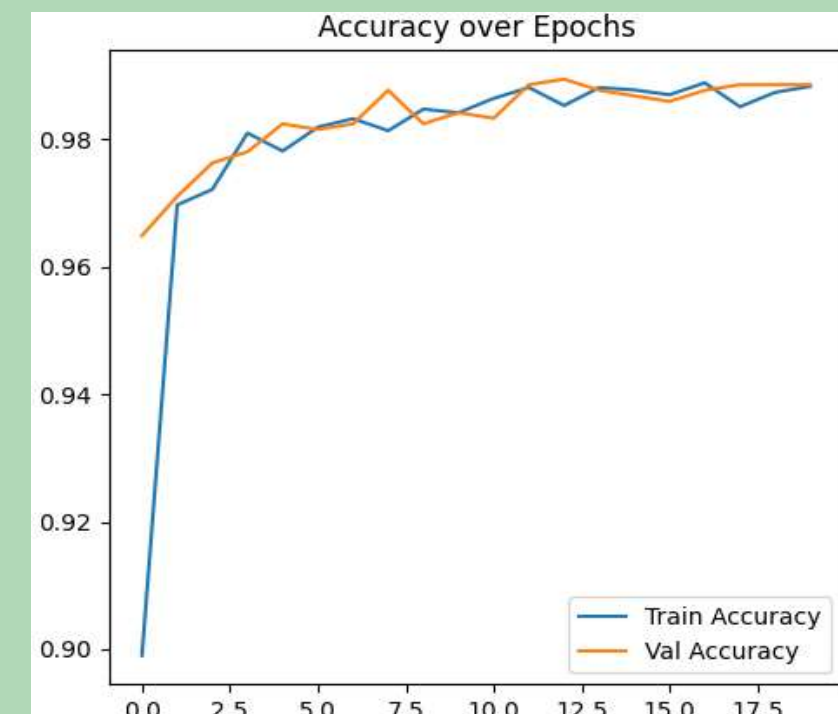
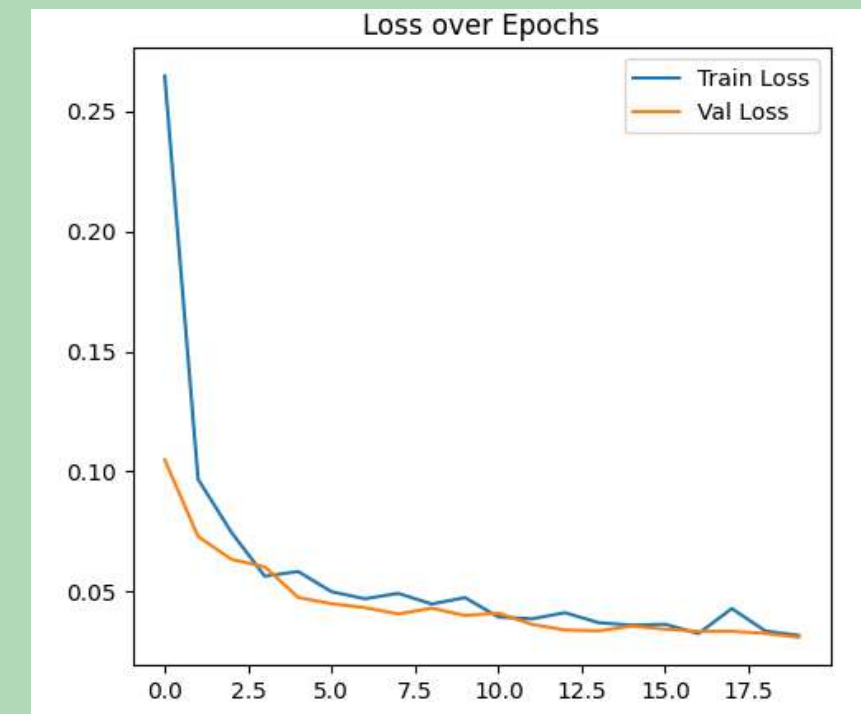
Methodology

- Collected dataset from Kaggle containing images of **masked and unmasked faces**.
- Applied **data preprocessing**:
 - Resizing (224×224), normalization, and augmentation (rotation, flip, brightness).
 - Built deep learning models using:
 - **CNN and MobileNetV2** for efficient and accurate detection.
 - Trained the model using:
 - Train (75%), Validation (15%), Test (10%)
 - Adam optimizer and binary cross-entropy loss
- Evaluated performance using:
 - Accuracy, Precision, Recall, F1 Score
- Implemented **real-time detection system**:
 - Input image → Preprocessing → Prediction → Output (Mask / No Mask)



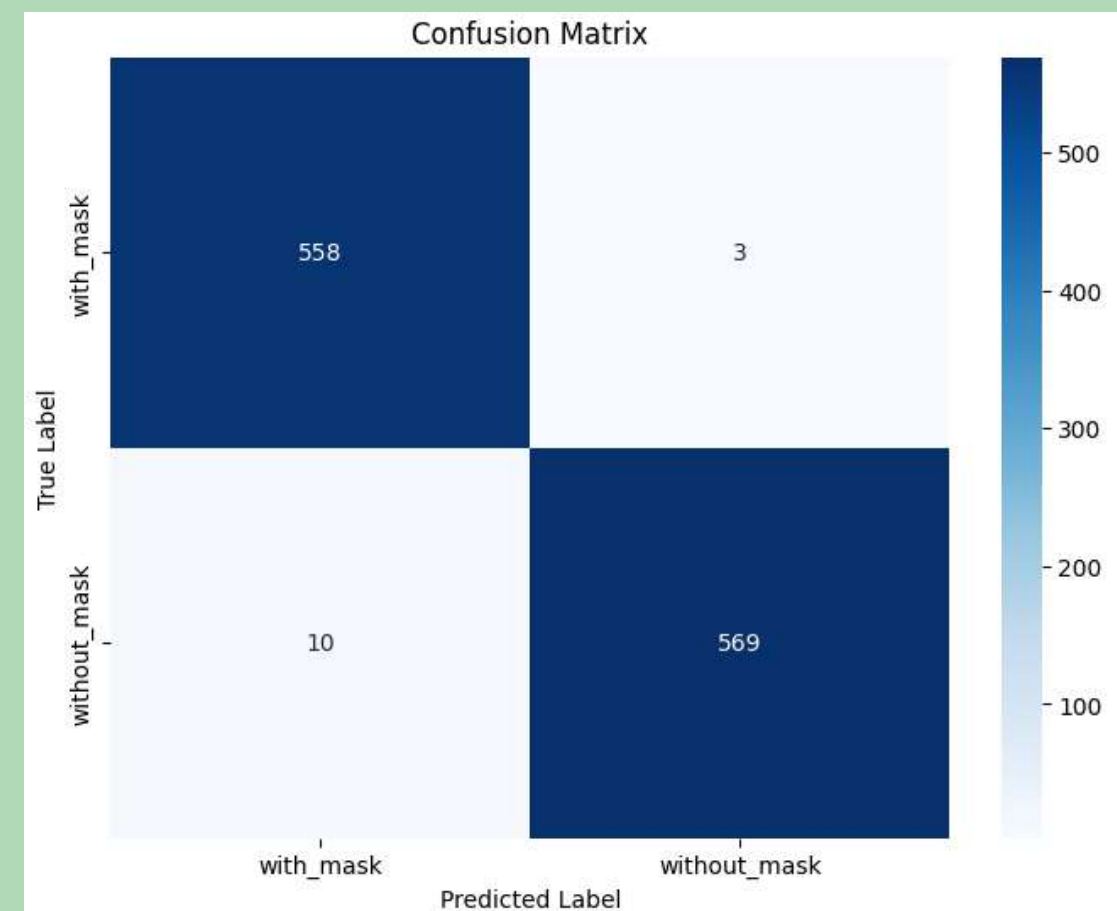
Results/discussion

- The proposed system achieved **high accuracy (83.2% to 100%)** across multiple datasets.
- Deep learning models performance:
- CNN: up to **99–100% accuracy**
- Transfer Learning: **98.7% – 99.64%**
- MobileNetV2: **93% – 94%**
- Provides **real-time detection** with very fast processing:
- Inference time < 3 ms per image**
- Model performs effectively under:
 - Different lighting conditions
 - Face orientations
 - Complex real-world environments
- Achieves **high precision and recall**, with very few misclassifications.
- Strengths:**
 - High accuracy and efficiency
 - Real-time performance
 - Suitable for smart surveillance and edge devices
- Limitations:**
 - Slight performance drop with:
 - Multiple faces
 - Occlusions
 - Improper mask usage
- Overall Impact:**
 - Enables automatic monitoring of mask compliance
 - Reduces manual effort
 - Improves public health safety
 - Suitable for real-world deployment



Conclusion

- The project successfully developed an **automatic face mask detection system** using deep learning techniques.
- The **MobileNetV2-based model** proved to be highly effective for accurate and efficient mask detection.
- The system achieved **high accuracy, precision, and F1-score**, showing strong performance compared to existing methods.
- Use of **data preprocessing and augmentation** improved model robustness and generalization.
- The model performs well on **unseen data and different real-world conditions**, including lighting and background variations.
- The system ensures **fast real-time detection** with low computational cost, making it suitable for practical deployment.
- It can be effectively used in:
 - Smart surveillance systems
 - Public places (airports, offices, malls)
 - Public health monitoring applications
- Overall Impact:**
 - Reduces manual monitoring efforts
 - Enhances public safety
 - Helps in controlling the spread of infectious diseases



Future Scope

- Detect improper mask usage (below nose/chin)
- Improve accuracy in crowded and complex environments
- Integrate with CCTV and smart surveillance systems
- Deploy on edge devices and IoT systems for real-time processing
- Use advanced deep learning models for better performance
- Train with larger and diverse datasets for better generalization
- Add alert/notification systems for violations
- Enable multi-class detection (mask / no mask / improper mask)
- Implement real-time video stream processing
- Develop mobile and cloud-based monitoring systems
- Improve performance in low-light conditions
- Optimize model for faster inference speed
- Support multi-camera systems and centralized monitoring
- Add AI-based analytics and reporting dashboard

References

- Al Amin et al., *Real-Time Face Mask Detection using CNN*, 2023
- Himeur et al., *Face Mask Detection using Deep & Transfer Learning*, 2023
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The background features a light green color with decorative circuit-like patterns in the corners. These patterns consist of dark blue lines and dots, some forming hexagonal grids and others resembling random circuit traces. Two horizontal dark blue lines with dots at their ends are positioned above and below the central text.

Thank You